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Math 102 Final Exam Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Which one of the following series is convergent?

(a) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[5]{n^3}}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^3}}$

(c) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[6]{n^3}}$

(d) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$

(e) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n^3}}$

2. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2+\sqrt{n}}$

(a) diverges

(b) diverges by the nth-term test for divergence

(c) converges by the ratio test

(d) converges conditionally

(e) converges absolutely

3. The interval of convergence I and the radius of convergence R of the power series $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 2^n \cdot x^n}{(n+1)(n+2)}$ are

(a) $I = (-\infty, \infty)$, $R = \infty$

(b) $I = \{0\}$, $R = 0$

(c) $I = \left[-\frac{1}{2}, \frac{1}{2}\right]$, $R = 1$

(d) $I = \left[-\frac{1}{2}, \frac{1}{2}\right]$, $R = \frac{1}{2}$

(e) $I = \left(-\frac{1}{2}, \frac{1}{2}\right)$, $R = \frac{1}{2}$

4. $\int_{3/2}^{9/4} \dfrac{1}{\sqrt{3x - x^2}}\, dx =$

(a) $\dfrac{2\pi}{3}$

(b) $\dfrac{\pi}{6}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{5\pi}{6}$

(e) $\dfrac{\pi}{3}$

5. The improper integral $\int_3^\infty \dfrac{1}{x\sqrt{x^2 - 9}}\, dx$

(a) converges to $\dfrac{\pi}{2}$

(b) converges to $2 \ln 3$

(c) converges to $\ln 3$

(d) diverges

(e) converges to $\dfrac{\pi}{6}$

6. $\int_2^3 \dfrac{-4x}{x^3 + x^2 - x - 1}\, dx =$

(a) $\ln\left(\dfrac{1}{6}\right) - \dfrac{2}{3}$

(b) $\ln\left(\dfrac{2}{3}\right) + \dfrac{3}{2}$

(c) $\ln\left(\dfrac{2}{3}\right) - \dfrac{1}{6}$

(d) $\ln\left(\dfrac{1}{6}\right) + \dfrac{1}{6}$

(e) $\ln\left(\dfrac{3}{2}\right) + \dfrac{1}{6}$

7. A particle moves along a line so that its velocity at time t is $v(t) = \cos t$ (in m/s). The total distance traveled by the particle during the time interval $0 \leq t \leq \pi$ is

(a) $\frac{1}{2}$

(b) 0

(c) 1

(d) 4

(e) 2

8. The volume of the solid generated by revolving the region bounded by the graphs of $y = \ln x$, $y = 0$ and $x = e$ about the x-axis is

(a) $\pi(2 + \ln 2)$

(b) $2\pi(2e - 1)$

(c) $\pi(e - 2)$

(d) $\pi(1 + \ln 2)$

(e) πe^2

9. The sequence $\left\{\frac{\cos(2n)}{3^n}\right\}$

(a) converges to $\frac{1}{3}$

(b) diverges

(c) converges to 0

(d) converges to 2

(e) converges to $\frac{2}{3}$

10. The series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3}$

(a) diverges by the integral test

(b) is a series for which the integral test is not applicable

(c) converges by the integral test

(d) diverges by the nth-term test for divergence

(e) is a convergent geometric series

11. The nth partial sum of the series $\sum_{n=1}^{\infty} \dfrac{4}{n(n+2)}$ is

(a) $S_n = \dfrac{n}{(n+1)(n+3)}$

(b) $S_n = \dfrac{2n+1}{(n+1)(n+2)}$

(c) $S_n = \dfrac{(n+2)(n+3)}{n(n+1)}$

(d) $S_n = \dfrac{n(3n+5)}{(n+1)(n+2)}$

(e) $S_n = \dfrac{3n+5}{n+2}$

12. $\displaystyle\int \frac{x + \ln\left(e^{2x-1}\right)}{2x}\, dx =$

(a) $\dfrac{1}{2}x + e^{2x-1} + C$

(b) $\dfrac{3}{2}x + \ln|x| + C$

(c) $\dfrac{1}{2}x - \dfrac{1}{2}\ln|2x-1| + C$

(d) $\dfrac{3}{2}x - \dfrac{1}{2}\ln|x| + C$

(e) $\dfrac{1}{2}x - \dfrac{1}{2x-1} + C$

13. $\int_{-3}^{0} \sqrt{9-x^2}\, dx =$

(a) $\dfrac{3\pi}{2}$

(b) $\dfrac{9\pi}{2}$

(c) $\dfrac{3\pi}{4}$

(d) $\dfrac{9\pi}{4}$

(e) $-\dfrac{9\pi}{4}$

14. The improper integral $\int_{-\infty}^{0} e^{3x}\,dx$

(a) converges to $3e$

(b) diverges

(c) converges to $\dfrac{1}{3}$

(d) converges to e^3

(e) converges to $-\dfrac{2}{3}$

15. $\sum_{n=2}^{\infty} \dfrac{(-1)^{n+1}\cdot 6}{(2n+1)!} =$

(a) $3 + 2\sin 1$

(b) $1 - 2\sin 1$

(c) $5 - 2\sin 1$

(d) $5 + 6\sin 1$

(e) $5 - 6\sin 1$

16. The series $\sum_{n=1}^{\infty} \dfrac{3\cdot 5\cdot 7\cdots(2n+1)}{18^n\cdot(2n-1)\cdot n!}$

(a) is a series with which the ratio test is inconclusive

(b) diverges by comparing it with $\sum_{n=1}^{\infty} \dfrac{1}{18^n}$

(c) converges by comparing it with $\sum_{n=1}^{\infty} \dfrac{1}{2n-1}$

(d) diverges by the ratio test

(e) converges by the ratio test

17. Let a be a real number such that $0 < a < 8$, and let

$$f(x) = \begin{cases} 4, & x < a \\ x, & x \geq a \end{cases}$$

If $\int_0^8 f(x)\,dx = 40$, then $2a - 5 =$

(a) 0

(b) -4

(c) 6

(d) -5

(e) 3

18. If $F(x) = \int_{\sqrt{x}}^{\sqrt{2}} 4\cos(t^4)\,dt$, then $F(2) + F'(2) =$

(a) $1 - \sqrt{2}\cos 4$

(b) $1 - \dfrac{\cos\sqrt{2}}{\sqrt{2}}$

(c) $\sqrt{2}\cos(2\sqrt{2})$

(d) $-\sqrt{2}\cos 4$

(e) $\dfrac{\cos 4}{\sqrt{2}}$

19. The series $\sum_{n=0}^{\infty}\left(\dfrac{2}{x}\right)^n$

(a) converges for $x < -2$ or $x > 2$ and its sum is $\dfrac{x}{x-2}$

(b) diverges for all values of x

(c) converges for $x < -1$ or $x > 1$ and its sum is $\dfrac{x}{x-1}$

(d) converges for $x \geq -2$ and its sum is $\dfrac{1}{x-2}$

(e) converges for $-2 < x < 2$ and its sum is $\dfrac{x}{x-2}$

20. A power series representation for $f(x) = \dfrac{4}{2+x^2}$, centered at 0, is given by

(a) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n-1}}{2^{n-1}}$

(b) $\sum_{n=0}^{\infty} \dfrac{x^{2n}}{2^{n-1}}$

(c) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^n}{2^n}$

(d) $\sum_{n=0}^{\infty} 2^{n+1}\cdot x^{2n}$

(e) $\sum_{n=0}^{\infty}(-1)^n \dfrac{x^{2n}}{2^{n-1}}$

21. The second Taylor polynomial for $f(x) = \sqrt{x}$ centered at $c = 4$ is

(a) $P_2(x) = 2 + \dfrac{1}{4}(x-4) - \dfrac{1}{32}(x-4)^2$

(b) $P_2(x) = 2 - \dfrac{1}{4}(x-4) - \dfrac{1}{64}(x-4)^2$

(c) $P_2(x) = 2 + \dfrac{1}{4}(x-4) - \dfrac{1}{64}(x-4)^2$

(d) $P_2(x) = 2 + (x-4) + (x-4)^2$

(e) $P_2(x) = 2 - \dfrac{1}{4}(x-4) + \dfrac{1}{32}(x-4)^2$

22. $\int_0^2 xe^{-2x^3}\,dx =$

(a) $\sum_{n=0}^{\infty}(-1)^n \dfrac{2^{3n+1}}{(3n+1)\cdot n!}$

(b) $\sum_{n=0}^{\infty}(-1)^n \dfrac{2^{3n+3}}{(3n+3)\cdot n!}$

(c) $\sum_{n=0}^{\infty}(-1)^n \dfrac{2^{3n+2}}{(3n+2)\cdot n!}$

(d) $\sum_{n=0}^{\infty}(-1)^n \dfrac{2^{4n+2}}{(3n+2)\cdot n!}$

(e) $\sum_{n=0}^{\infty}(-1)^n \dfrac{2^{4n+1}}{(3n+1)\cdot n!}$

23. Which one of the following statement is FALSE?

(a) If $0 \le a_n \le \dfrac{1}{n^2}$, $n \ge 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent

(b) If $a_n \ge 2^n$, $n \ge 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent

(c) If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty}\left(a_n + \dfrac{1}{n}\right)$ is convergent

(d) If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty} 2a_n$ is divergent

(e) If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty}\left(a_n + \dfrac{1}{n}\right)$ is divergent

24. The area of the region enclosed by the curves $y = \sqrt{x+1}$, $y = 1 - x$, and $y = 0$ is

(a) $\dfrac{5}{6}$

(b) $\dfrac{1}{6}$

(c) $\dfrac{7}{6}$

(d) $\dfrac{13}{6}$

(e) $\dfrac{11}{6}$

25. Which one of the following series is convergent?

(a) $\sum_{n=1}^{\infty}(-1)^n$

(b) $\sum_{n=1}^{\infty}(-1)^n\dfrac{n}{n+1}$

(c) $\sum_{n=1}^{\infty}(-1)^n\cos\left(\dfrac{1}{n}\right)$

(d) $\sum_{n=1}^{\infty}(-1)^n\sqrt{n}$

(e) $\sum_{n=1}^{\infty}(-1)^n\sin\left(\dfrac{1}{n}\right)$

26. The volume of the solid generated by rotating the region bounded by the curves $y = x^2 + 1$, $y = 0$, $x = 0$ and $x = 1$ about the y-axis is

(a) $\dfrac{3\pi}{4}$

(b) $\dfrac{3\pi}{2}$

(c) 2π

(d) $\dfrac{\pi}{2}$

(e) $\dfrac{\pi}{4}$

27. $\displaystyle\int \frac{2x}{(x+1)^2}\,dx =$

(a) $\dfrac{1}{2}\ln|x+1| + 2(x+1) + C$

(b) $2x - \arctan x + C$

(c) $\dfrac{1}{(x+1)^2} + \dfrac{2}{x+1} + C$

(d) $-2x + \ln|x+1| + C$

(e) $2\ln|x+1| + \dfrac{2}{x+1} + C$

28. $\displaystyle\int_{\ln 2}^{\ln(2/\sqrt{3})} \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,dx =$

(a) $-\dfrac{\pi}{6}$

(b) $\dfrac{\pi}{6}$

(c) $-\dfrac{\pi}{3}$

(d) $\dfrac{\pi}{3}$

(e) π

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	p-series: exponents $\frac{3}{2}, 1, \frac{3}{4}, \frac{3}{5}, \frac{1}{2}$; only $p = \frac{3}{2} > 1$ converges.
2	**(d)**	AST: converges. LCT with $\sum n^{-1/2}$: $\sum \lvert a_n \rvert$ diverges. Conditional.
3	**(d)**	Ratio test: $R = \frac{1}{2}$; both endpoints converge, so $I = [-\frac{1}{2}, \frac{1}{2}]$.
4	**(b)**	Complete the square: $\left[\arcsin \frac{2u}{3}\right]_0^{3/4} = \frac{\pi}{6}$.
5	**(e)**	$\frac{1}{3} \operatorname{arcsec} \frac{x}{3}\big\rvert_3^\infty = \frac{\pi}{6}$: converges.
6	**(c)**	$\frac{-1}{x-1} + \frac{1}{x+1} - \frac{2}{(x+1)^2}$: value $\ln \frac{2}{3} - \frac{1}{6}$.
7	**(e)**	$\int_0^\pi \lvert \cos t \rvert dt = 1 + 1 = 2$.
8	**(c)**	Disks: $\pi \int_1^e (\ln x)^2 dx = \pi[x \ln^2 x - 2x \ln x + 2x]_1^e = \pi(e-2)$.
9	**(c)**	$\lvert \cos 2n \rvert / 3^n \le 3^{-n} \to 0$: squeeze gives limit 0.
10	**(c)**	$\int_2^\infty \frac{dx}{x \ln^3 x} = \frac{1}{2(\ln 2)^2} < \infty$: converges.
11	**(d)**	Telescoping: $S_n = 2[\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2}] = \frac{n(3n+5)}{(n+1)(n+2)}$.
12	**(d)**	$\frac{3x-1}{2x} = \frac{3}{2} - \frac{1}{2x}$, giving $\frac{3}{2}x - \frac{1}{2} \ln \lvert x \rvert + C$.
13	**(d)**	Quarter disk of radius 3: $\frac{1}{4}\pi \cdot 9 = \frac{9\pi}{4}$.
14	**(c)**	$\lim_{t \to -\infty} \frac{1}{3}(1 - e^{3t}) = \frac{1}{3}$: converges.

Q	Ans	Brief explanation
15	**(e)**	$-6(\sin 1 - 1 + \frac{1}{6}) = 5 - 6\sin 1$.
16	**(e)**	$\frac{a_{n+1}}{a_n} = \frac{(2n+3)(2n-1)}{18(2n+1)(n+1)} \to \frac{1}{9} < 1$: converges.
17	**(e)**	$4a + \frac{64-a^2}{2} = 40 \Rightarrow (a-4)^2 = 0$, $a = 4$, so $2a - 5 = 3$.
18	**(d)**	$F(2) = 0$, $F'(x) = -\frac{2\cos x^2}{\sqrt{x}}$, so the sum is $-\sqrt{2}\cos 4$.
19	**(a)**	Geometric, $r = 2/x$: converges for $\lvert x\rvert > 2$, sum $\frac{1}{1-2/x} = \frac{x}{x-2}$.
20	**(e)**	$\frac{2}{1+x^2/2} = \sum(-1)^n \frac{x^{2n}}{2^{n-1}}$, $\lvert x\rvert < \sqrt{2}$.
21	**(c)**	$f(4) = 2$, $f'(4) = \frac{1}{4}$, $\frac{f''(4)}{2} = -\frac{1}{64}$.
22	**(d)**	$e^{-2x^3} = \sum \frac{(-2x^3)^n}{n!}$, integrate termwise: $\sum(-1)^n \frac{2^{4n+2}}{(3n+2)n!}$.
23	**(c)**	Counterexample $a_n = \frac{1}{n}$: $\sum a_n$ and $\sum(a_n + \frac{1}{n})$ both diverge, so the first claim is false.
24	**(c)**	Split at $x = 0$: $\frac{2}{3} + \frac{1}{2} = \frac{7}{6}$.
25	**(e)**	Only $\sin(1/n)$ decreases to 0: alternating series test; the others have terms not tending to 0.
26	**(b)**	Shells: $2\pi \int_0^1 (x^3 + x)dx = \frac{3\pi}{2}$.
27	**(e)**	$u = x + 1$: $2\int(\frac{1}{u} - \frac{1}{u^2})du = 2\ln\lvert x+1\rvert + \frac{2}{x+1} + C$.
28	**(a)**	$u = e^{-x}$: $-[\arcsin u]_{1/2}^{\sqrt{3}/2} = -\frac{\pi}{6}$.